Assignment: VLM

Develop an AI-based system that takes an animal image and a natural-language prompt as input, identifies and segments the specified body part, displays the isolated body part, and provides a short description of its appearance and biological function.

Tasks

Part A: System Development Without a Vision-Language Model (VLM)

1. Develop a system using conventional computer vision and deep learning models to identify and segment animal body parts based on text prompts.
2. Integrate suitable object-detection and image-segmentation models, such as YOLO and SAM 2, to locate and isolate the specified body part.
3. Generate a short description of the segmented body part using predefined descriptions or a language model that does not use visual input.
4. Display the isolated body part alongside its description through an interactive user interface.

Part B: System Development Using a Vision-Language Model (VLM)

1. Develop a system that integrates a pretrained VLM with suitable object-localization and image-segmentation models.
2. Allow users to upload an animal image and enter natural-language prompts, such as *"Show the wings of this bird"* or *"Identify the ears of this cat."*
3. Use vision-language understanding to interpret the prompt and identify the requested body part, then segment it using an appropriate segmentation model.
4. Display the isolated body part and use a VLM to generate a short, context-aware description of its appearance and biological function.

Part C: Comparative Evaluation

1. Test both systems on different animal species and body parts using the same set of images and prompts.
2. Compare their segmentation quality, prompt-following accuracy, description quality, and computational requirements.
3. Evaluate segmentation performance using appropriate metrics, such as Intersection over Union (IoU) and Dice coefficient, where ground-truth masks are available.
4. Discuss the advantages, limitations, and suitability of each approach for prompt-guided animal body-part segmentation and description.

Deliverables

- Source code and a functional implementation of both systems uploaded in GitHub.

- Sample input images and text prompts.
- Segmented body-part images and generated descriptions from both approaches.
- Comparative experimental results and performance analysis.
- A technical report documenting the methodology, model architectures, experimental setup, results, discussion, and conclusions.

Expected outcome: A comparative study demonstrating whether integrating a VLM improves prompt interpretation, body-part identification, and description generation compared with a system based on conventional computer vision and deep learning models.